Roll No. | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS-THEORY

Subject Code : BP-604T

M.Code : 77989

Date of Examination : 10-01-23

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.**

SECTION-A

1. Write briefly :

a) What do you mean by hepatic first pass metabolism.

b) What is active secretion?

c) What is absolute bioavailability?

d) Mention the formula for calculating T1/2 of drug for first order kinetics.

e) What is extraction ratio?

f) What is inulin clearance?

g) Give two examples of plasma proteins.

h) Define the properties of drugs belonging to BCS Class I & II.

i) What is the unit of Ka?

j) Mention the formula for calculating Vd from dose and Cp.

SECTION-B

2. Give a detailed account of the physiological factors influencing drug absorption.

3. IV bolus dose (4 mg / Kg) was administered to a patient of 75 kg weight. Following equation represented the drug kinetics:

   $C_p = 78\ e^{-0.46t}$ Calculate: (a) T1/2; (b) Vd; (c) Plasma concentration after 4 hr.

4. What is meant by non-linear pharmacokinetics? Discuss the factors responsible for non-linear pharmacokinetics of drugs.

SECTION-C

5. Write briefly about facilitated and active transport of drugs.

6. Explain with the help of suitably equations the pharmacokinetics of a drug in plasma after IV administration that follows one compartment open model.

7. Discuss the regulatory considerations pertaining to bioequivalence studies in India.

8. Write briefly about protein binding of drugs.

9. What is Sigma-Minus method? Explain the method of calculating elimination rate constant by this method with the help of suitable equations.

10. Discuss the role of pKa of drug and pH of biological fluid in drug absorption.

11. Discuss the phase -1 reactions for drug metabolism.

12. Write a note on the approaches used for enhancing solubility and dissolution rates of poorly water soluble drugs.

13. Comment on in vitro - in vivo correlations.

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.